The Verification Boundary — summary brief

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ABSTRACT

Strategy and organization theory explain why firms integrate complementary assets, envelop adjacent markets, and select governance that economizes on transaction costs. They do not specify the recurring limit that appears when an automated system produces consequential outputs and a third party must rely on an assurance about them. This work develops the **verification boundary**, a stopping rule that platform envelopment leaves open. The scarce asset is not superior intelligence, nor even superior error detection; it is credible independence.

KEY CLAIM

> *When an automated output can create material, difficult-to-recover losses for parties beyond its producer, and an external actor conditions money, permission, admissibility, or liability on an assurance about that output, credible verification must remain institutionally independent from generation. Improvements in the generator's technical capability do not, by themselves, relax this requirement.*

Scope conditions are conjunctive: **(i)** consequential output, **(ii)** reliance by a party other than the producer, **(iii)** an external actor conditioning a decision on the assurance. Independence admits four forms, by strength: protected internal function, ring-fenced subsidiary, regulated professional role, separate firm. Separate-firm formation is a contingent response, not the proposition. The boundary sits at Layer 3 (Gatekeeping) of the ten-layer framework, which is context, not a premise. Positioning: Eisenmann, Parker & Van Alstyne (2011); Teece (1986); Coase (1937); Williamson (1985); Akerlof (1970); Dulleck & Kerschbamer (2006); Power (1997).

FALSIFIABLE PREDICTIONS (FIVE-YEAR HORIZON)

	Expected observation	Refuted if
P1 Separation	Accepted assurance flows through a protected function, ring-fenced entity, regulated role, or separate firm.	Conditioning parties routinely accept generator assurance with no separation.
P2 Capability	As accuracy rises, verification automates and narrows; independent accountability persists.	Higher accuracy systematically causes assurance requirements to be dropped.
P3 Expansion bends	Platforms expand into workflow, access, and surface more readily than into assurance of their own outputs.	A platform repeatedly absorbs such assurance and keeps third-party acceptance.

	Expected observation	Refuted if
P4 Private demand	Separation appears wherever a private payer conditions money or liability, absent any mandate.	Separation appears only under statute.
P5 Acquisition	Acquired assurance providers are ring-fenced or lose third-party acceptance.	Integrated providers keep acceptance and pricing power over multiple cycles.
P6 New mandates	New requirements produce an independent role or accreditation within roughly two years.	Newly regulated domains resolve into single-vendor generation-plus-verification.

Whole-claim refutation. One well-documented market in which all three scope conditions hold and the generator's own attestation is durably accepted by the conditioning third party, across multiple cycles and without ring-fencing.

THE ONE QUESTION ASKED OF REVIEWERS

Is the verification boundary already contained in an existing result under another name? If it is, a citation is the most useful reply. If it is not, the sharpest contribution is a counter-example satisfying all three scope conditions that survives the test above.

HOW TO CITE

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